MATH 440/540 **HW 13** Schreyer

Due Thurs. Nov. 14

WITH SOLUTIONS

This material will be on the test that will be given on Wed. Nov. 20. Solutions will be posted on Mon. Nov. 18 or sooner.

1. (16 pts) Z& T Chapter 8, 2.4. Consider

$$\begin{aligned} &u_{xx} - u_{tt} = 0, \qquad x \in I\!R, \quad t > 0 \\ &u(x,0) = \phi(x) \qquad u_t(x,0) = \psi(x), \qquad x \in I\!R \end{aligned}$$

For each of the following ϕ and ψ, find the d'Alembert solution, evaluate your solution at $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$ for $t = 0,\ \frac{\pi}{2},\ \pi, \frac{3\pi}{2},\ 2\pi$, and **sketch, by hand** the solution for 4 or 5 time steps. The sketch should just give the idea of the solution and does not need to be precise - be sure $t = 0$ is correct, and that your amplitudes are correct.

(a)

$$\begin{aligned} &u(x,0) = \phi(x) = \begin{cases} \cos(x) & \text{for} \quad |x| < \frac{\pi}{2} \\ 0 & \text{for} \quad |x| \geq \frac{\pi}{2} \end{cases} \\ &u_t(x,0) = \psi(x) = 0, \qquad x \in I\!R \end{aligned}$$

(b)

$$\begin{aligned} &u(x,0) = \phi(x) = 0, \qquad x \in I\!R \\ &u_t(x,0) = \psi(x) = \begin{cases} \cos(x) & \text{for} \quad |x| < \frac{\pi}{2} \\ 0 & \text{for} \quad |x| \geq \frac{\pi}{2} \end{cases} \end{aligned}$$

Grader: For sketches - check $t = 0$ and amplitudes, and general trends only.

(a) The d'Alembert's solution is:

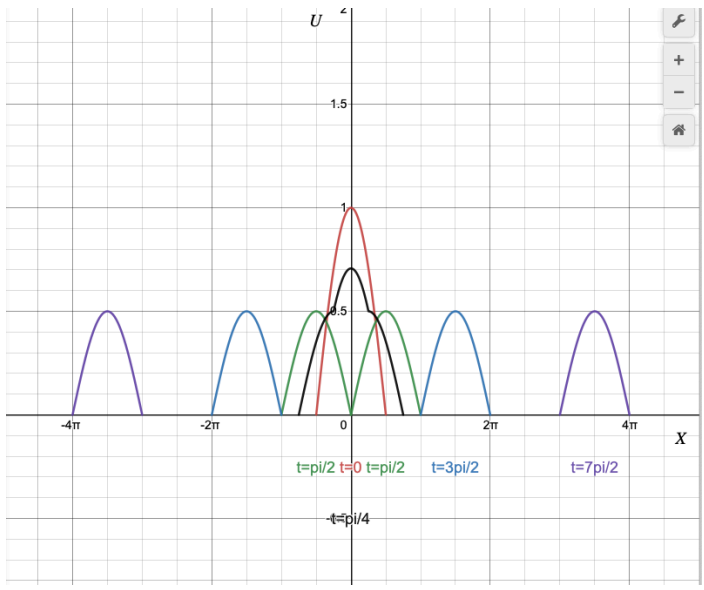
$$u(x,t) = \frac{1}{2}\left[\phi(x+t) + \phi(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} \psi(\tau)\, d\tau$$

With $\psi(x) = 0$ we get

$$u(x,t) \quad = \quad \frac{1}{2}\left[\phi(x+t) + \phi(x-t)\right]$$

Using this solution we can plug in values for x and t to get the displacement at any point. For example let us choose the point $t = \pi/4$ and $x = \pi/2$. Then $x + t = 3\pi/2$ which is outside of the interval, $(-\pi/2, \pi/2)$, so $\phi(x+t) = 0$. Likewise, $\phi(x-t) = \phi(\pi/4) = cos(\pi/4) = \sqrt{2}/2$. So we have $u(\pi/2, \pi/4) =$

$1/2(0+\sqrt{2}/2) = \sqrt{2}/4$. Doing the same for the other points we get:
$u(\pi,0) = 0$, $u(\pi,\pi/2) = 0$, $u(\pi,\pi) = \dfrac{1}{2}$, $u(\pi,3\pi/2) = 0$, $u(\pi,2\pi) = 0$,
$u(-\pi,0) = 0$, $u(-\pi,\pi/2) = 0$, $u(-\pi,\pi) = \dfrac{1}{2}$, $u(-\pi,3\pi/2) = 0$, $u(-\pi,2\pi) = 0$.
Using the fact that $x+t$ is a plane wave moving left and $x-t$ is the plane wave moving right, we obtain the following (using Desmos). Note that the graph is consistent with the values of calculated above.



(b) Switching the initial conditions, we now have $\phi = 0$ and our d'Alembert solution is

$$u(x,t) = \frac{1}{2}\int_{x-t}^{x+t} \psi(x')\, dx'$$

Using this solution we can plug in values for x and t to get the displacement at any point. For example let us choose the point $t = \pi/4$ and $x = \pi/2$. Then $x+t = 3\pi/2$ and $x-t = \pi/4$. So we have

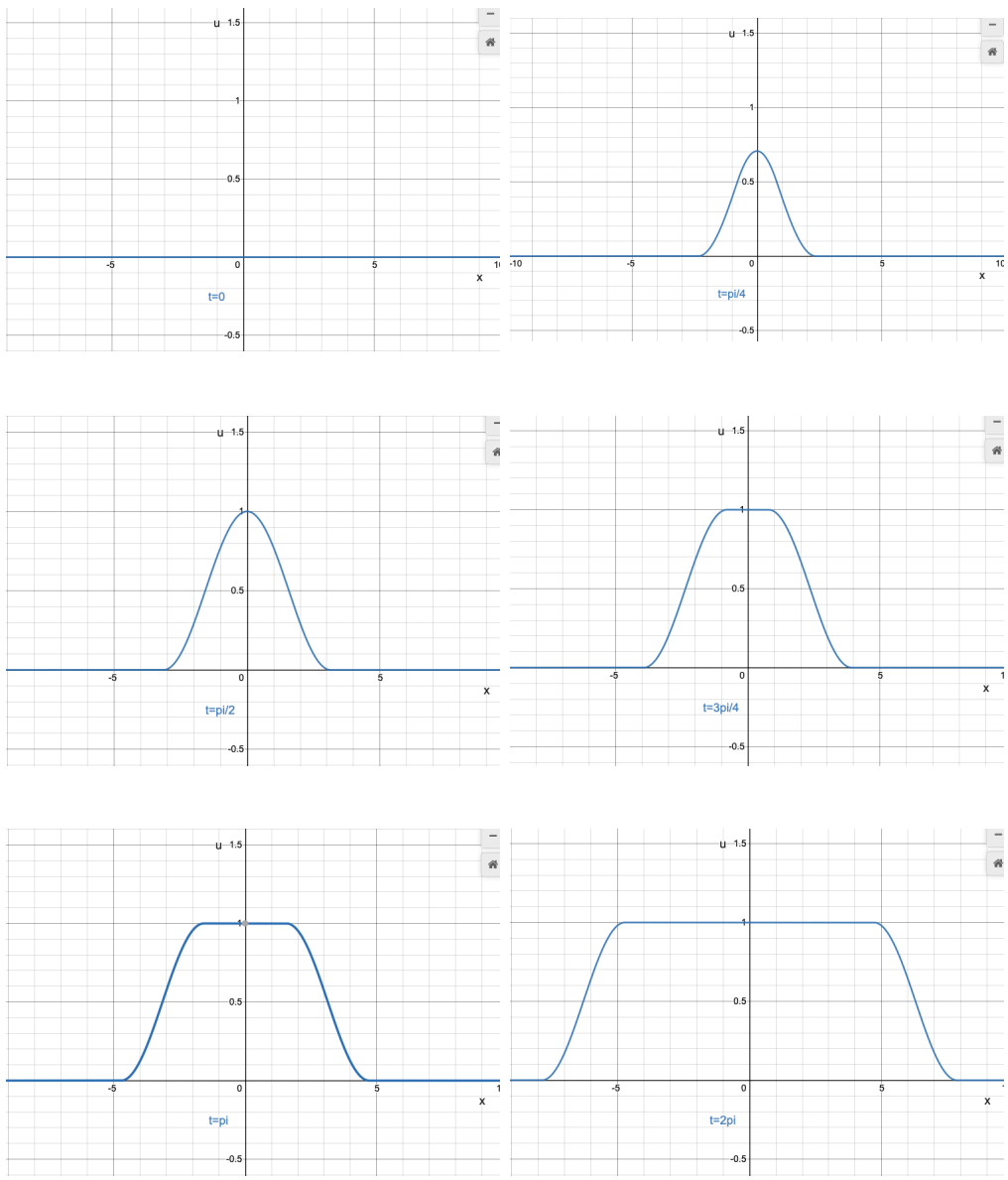
$$u(\pi/2,\pi/4) = \frac{1}{2}\int_{\pi/4}^{3\pi/4} \psi(x')\, dx' = \frac{1}{2}\int_{\pi/4}^{\pi/2} \cos(x')\, dx' + \frac{1}{2}\int_{\pi/2}^{3\pi/4} 0\, dx' = \frac{\sqrt{2}}{4}.$$

Repeating this sort of calculation we get:
$u(\pi,0) = 0$, $u(\pi,\pi/2) = 0$, $u(\pi,\pi) = \dfrac{1}{2}$, $u(\pi,3\pi/2) = 1$, $u(\pi,2\pi) = 1$,
$u(-\pi,0) = 0$, $u(-\pi,\pi/2) = 0$, $u(-\pi,\pi) = \dfrac{1}{2}$, $u(-\pi,3\pi/2) = 1$, $u(-\pi,2\pi) = 1$.
This is not so easy to graph. Here are some clues:

- At $t = 0$ $u = 0$ (everywhere).
- Since $\psi(x') \geq 0$ then $0 \geq u(x,t)$, i.e. u can never be negative.

- When $x - t_0 < -\frac{\pi}{2}$ and $x + t_0 > \frac{\pi}{2}$ then the solution is constant for all $t > t_0$, and $u = \frac{1}{2}\int_{-\pi/2}^{\pi/2} \cos(x')\, dx' = 1$, which is the biggest it can be. This occurs when $\frac{\pi}{2} - t_0 < x < t_0 - \frac{\pi}{2}$.
- When $x - t_0 > -\frac{\pi}{2}$ or $x + t_0 < \frac{\pi}{2}$ then we are not integrating over the whole region where $\psi(x')$ is not zero, so we do not get to $u = 1$ and $0 < u < 1$.

Using these observations and the check from above we get:



2. (4 pts) Z& T Chapter 8, 3.1 Verify the following identity:

$$2u_t\left(u_{x_1x_1} + u_{x_2x_2} - u_{tt}\right) = (2u_tu_{x_1})_{x_1} + (2u_tu_{x_2})_{x_2} - (u_{x_1}^2 + u_{x_2}^2 + u_t^2)_t$$

Method 1: Start from RHS and go to LHS.

$$
\begin{aligned}
&(2u_t u_{x_1})_{x_1} + (2u_t u_{x_2})_{x_2} - (u_{x_1}^2 + u_{x_2}^2 + u_t^2)_t \\
&\quad = 2u_{tx_1}u_{x_1} + 2u_t u_{x_1x_1} + 2u_{tx_2}u_{x_2} + 2u_t u_{x_2x_2} - 2u_{x_1}u_{x_1t} - 2u_{x_2}u_{x_2t} - 2u_t u_{tt} \\
&\quad = 2u_t u_{x_1x_1} + 2u_t u_{x_2x_2} - 2u_t u_{tt} \\
&\quad = 2u_t\left(u_{x_1x_1} + u_{x_2x_2} - u_{tt}\right)
\end{aligned}
$$

Method 2: Starting from LHS and going to RHS (this is hard without doing Method 1 first)

$$
\begin{aligned}
2u_t\left(u_{x_1x_1} + u_{x_2x_2} - u_{tt}\right) &= 2u_t u_{x_1x_1} + 2u_t u_{x_2x_2} - 2u_t u_{tt} \\
&= (2u_t u_{x_1})_{x_1} - 2u_{tx_1}u_{x_1} + (2u_t u_{x_2})_{x_2} - 2u_{tx_2}u_{x_2} - 2u_t u_{tt} \\
&= (2u_t u_{x_1})_{x_1} - (u_{x_1})_t^2 + (2u_t u_{x_2})_{x_2} - (u_{x_2})_t^2 u_{x_2} - (u_t)_t^2 \\
&= (2u_t u_{x_1})_{x_1} + (2u_t u_{x_2})_{x_2} - (u_{x_1}^2 + u_{x_2}^2 + u_t^2)_t
\end{aligned}
$$

3. (6 pts) Z& T Chapter 8, 4.1. Suppose that u is a C^2 solution of the wave equation

$$\frac{\partial^2 u}{\partial x_1^2} + \ldots + \frac{\partial^2 u}{\partial x_1^n} - \frac{\partial^2 u}{\partial t^2} = 0, \qquad x \in I\!R^n, \quad t > 0,$$

and that at $t = 0$, $u = u_t = 0$ in a ball of radius R. Up to what time t can you be sure that $u = 0$ at the center of the ball? (Consider solving geometrically).

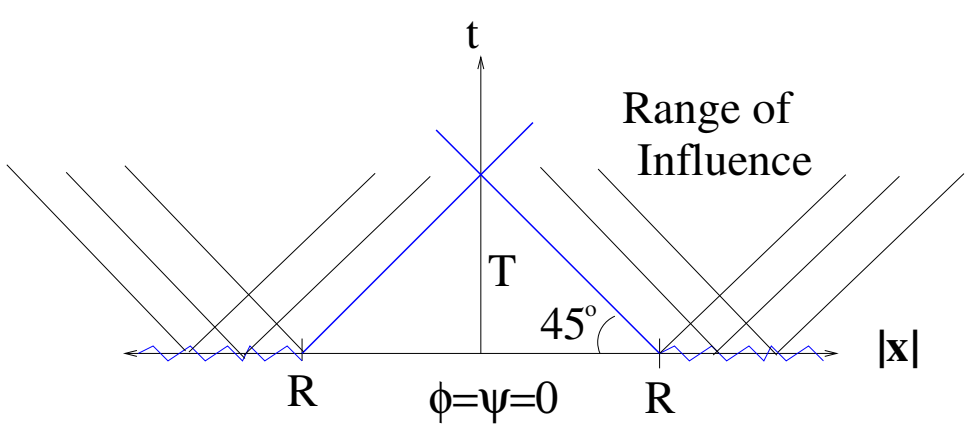
We consider the initial value problem for the n-dimensional wave equation:

$$
\begin{aligned}
&\triangle u - \frac{1}{c^2}u_{tt} = 0, \qquad x \in I\!R^n, \quad t > 0 \\
&u(x,0) = \phi(x) \qquad u_t(x,0) = \psi(x), \qquad x \in I\!R^n
\end{aligned}
\tag{1}
$$

Let us assume the data, (ϕ, ψ) are zero in the ball centered at $\boldsymbol{x}_0$ with radius R. We wish to know at what time, T, we are no longer guaranteed that the solution at $\boldsymbol{x}_0$ is no longer zero.

Since the speed of the wave is given by $c = 1$, the range of influence are formed by cones with angles of 45 degrees. Using the figure below, we solve:

$$\tan(45^\circ) = \frac{T}{R} \quad \Rightarrow \quad T = R.$$

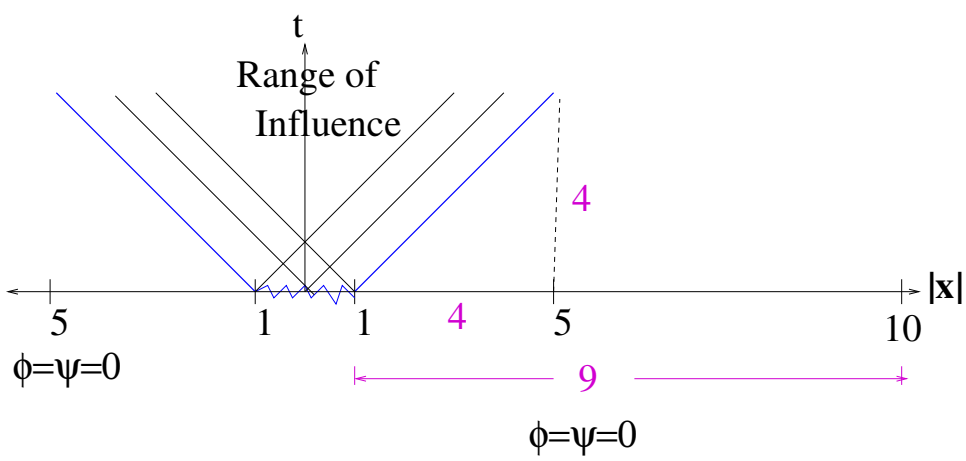


4. (6 pts) Z& T Chapter 8, 4.2. Suppose that u is a C^2 solution of the two-dimensional wave equation and that at $t = 0$, $u = u_t = 0$ outside the disc $x_1^2 + x_2^2 \leq 1$. Up to what time can you be sure that $u = 0$ at the points:
(a) $(x_1, x_2) = (5, 0)$ (b) $(x_1, x_2) = (0, 10)$ (c) $(x_1, x_2) = (2, 3)$?
(Consider solving geometrically).

We still consider the IVP (**??**), but now the support of the data is contained in the ball centered at the origin with radius 1. We want to know at what time, t, for which we are no longer guaranteed the solution is zero at the values of x given below. We use the range of influence and the sketch below to see we want to find t where

$$\tan(45^\circ) = \frac{t}{|\boldsymbol{x}| - 1} = 1$$

- $|\boldsymbol{x}| = 5 \quad \Rightarrow \quad t = 5 - 1 = 4$
- $|\boldsymbol{x}| = 10 \quad \Rightarrow \quad t = 10 - 1 = 9$
- $|\boldsymbol{x}| = 4 \quad \Rightarrow \quad t = \sqrt{4+9} - 1 = \sqrt{13} - 1.$



5. (6 pts) Z& T Chapter 8, 4.5. Show that if u is the solution of the initial value problem

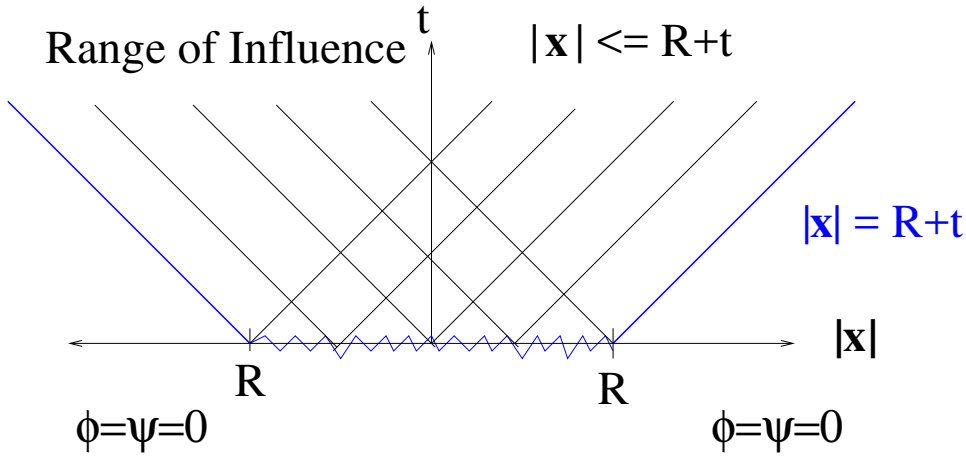
$$\frac{\partial^2 u}{\partial x_1^2} + ... + \frac{\partial^2 u}{\partial x_1^n} - \frac{\partial^2 u}{\partial t^2} = 0, \qquad x \in I\!R^n, \quad t > 0,$$
$$u(\boldsymbol{x}, 0) = \phi(\boldsymbol{x}), \qquad u_t(\boldsymbol{x}, 0) = \psi(\boldsymbol{x}), \qquad \boldsymbol{x} \in I\!R$$

and if the initial data ϕ and ψ vanish outside the ball $\overline{B}(0, R)$ in $I\!R^n$, then $u(x, t)$ must vanish outside the set of points defined by the inequalities

$$r \leq R + t, \qquad t \geq 0, \tag{2}$$

where $r = |\boldsymbol{x}|$. The set described above, (1), may be called the range of influence of the data in the ball $\overline{B}(0, R)$. Sketch the set described by (1) above for $n = 1$ and $n = 2$.

By range of influence and using the sketch below, $u = 0$ for $t < |\boldsymbol{x}| - R$ or $r = |\boldsymbol{x}| < R + t$. The sketches are below.



6. (6 pts) Z& T Chapter 8, 5.5 [don't forget c and do not assume c is 1.] Consider the initial value problem

$$\frac{\partial^2 u}{\partial x_1^2} + ... + \frac{\partial^2 u}{\partial x_n^2} - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = 0, \qquad \boldsymbol{x} \in I\!R^n, \quad t > 0$$
$$u(x,0) = \phi(x), \quad \frac{\partial u}{\partial t}(x,0) = \psi(x), \qquad \boldsymbol{x} \in I\!R^n$$

Write down the formulas for the solution of this problem for $n = 2$ and $n = 3$. Recall that the one-dimensional case was done in Chapter 8, problem 2.2.

For $n = 1$ the solution was found in problem 2 (Z& T, Chapter 8, problem 2.2) above. Using the same change of variables, $t' = ct$ and for the same reason replacing the initial velocity, $\psi(x)$, with $\frac{1}{c}\psi(x)$, we get

<u>n=2</u>

$$\begin{aligned} u(x_1,x_2,t) &= \frac{1}{2\pi c}\int_{\overline{B}(x_1,x_2;ct)} \frac{\psi(x_1',x_2')}{\sqrt{c^2t^2-(x_1'-x_1)^2-(x_2'-x_2)^2}}\,dx_1'dx_2' \\ &\quad + \frac{1}{2\pi c}\frac{\partial}{\partial t}\int_{\overline{B}(x_1,x_2;ct)} \frac{\phi(x_1',x_2')}{\sqrt{c^2t^2-(x_1'-x_1)^2-(x_2'-x_2)^2}}\,dx_1'dx_2' \end{aligned}$$

<u>n=3</u>

$$u(\boldsymbol{x},t) = \frac{1}{4\pi c^2 t}\int_{S(\boldsymbol{x},ct)} \psi(\boldsymbol{x}')\,d\sigma_{ct} + \frac{\partial}{\partial t}\left[\frac{1}{4\pi c^2 t}\int_{S(\boldsymbol{x};ct)} \phi(\boldsymbol{x}')\,d\sigma_{ct}\right]$$

7. (16 pts) Determine whether the following problems are well-posed. Assume u is as smooth as is necessary and that Ω is an appropriate domain. Respond with A, B, or C. If C explain why:
(A) Guaranteed to be well-posed
(B) "Well-posed" but solution is only unique up to a constant.
(C) Not guaranteed to be well-posed. Explain.

(a)

$$\begin{aligned} &\triangle u = 0 \qquad 0 \le r < 4, \quad -\pi \le \theta \le \pi \\ &u(4,\theta) = |\theta| \qquad -\pi \le \theta \le \pi \end{aligned}$$

(A)

(b)

$$\begin{aligned} &\triangle u = 0 \qquad 0 \le r < 4, \quad -\pi \le \theta \le \pi \\ &\frac{\partial u}{\partial r}(4,\theta) = 3 \qquad -\pi \le \theta \le \pi \end{aligned}$$

(C) Net flux out of the domain is not zero - no steady-state solution.

(c)

$$\begin{aligned} &\triangle u = 0 \qquad 0 < x < 1, \quad 0 < y < 1 \\ &\frac{\partial u}{\partial n}(x,y) = \begin{cases} 0 & 0 \le x \le 1, \quad y = 0 \\ -1 & 0 \le x \le 1, \quad y = 1 \\ 0 & 0 \le y \le 1, \quad x = 0 \\ 1 & 0 \le y \le 1, \quad x = 1 \end{cases} \end{aligned}$$

where $\boldsymbol{n}$ is the unit vector out of the square domain.

(C) The net flux out of the domain is zero - but it is not continuous, so we are not guaranteed the problem is well-posed.

(d)

$$\begin{aligned} &\triangle u - u_{tt} = 0 \qquad \boldsymbol{x} \in I\!R^n, \quad t > 0 \\ &u(\boldsymbol{x},0) = \begin{cases} \cos(\boldsymbol{x}), & |\boldsymbol{x}| \le \frac{\pi}{2} \\ 0 & |\boldsymbol{x}| > \frac{\pi}{2} \end{cases} \end{aligned}$$

(C) The IVP for the wave equation needs two initial conditions.